Assignment 11:

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Q1–Code:

main.cpp

```cpp
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

int matrixChainOrder(const vector<int>& p) {
    int n = p.size();
    vector<vector<int>> dp(n, vector<int>(n, 0));

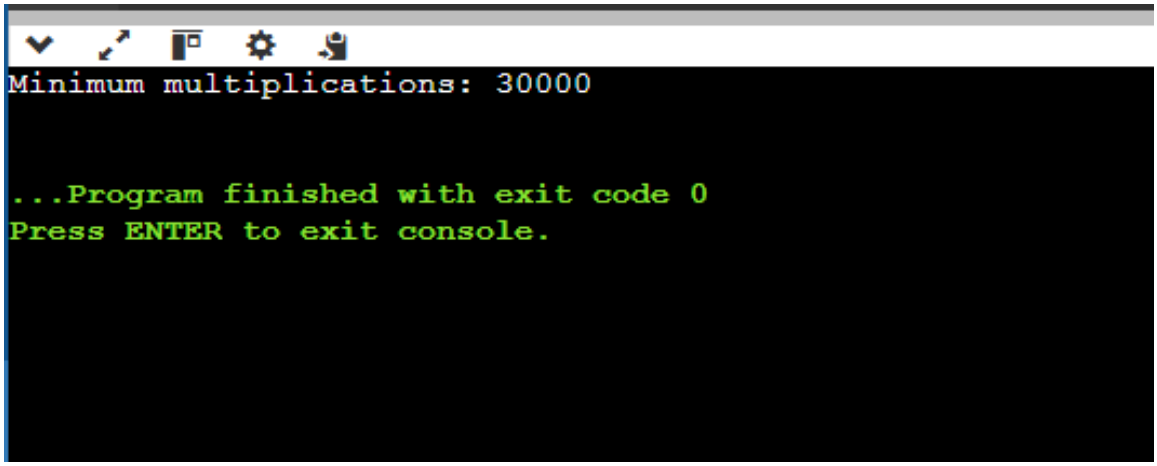
    for (int len = 2; len < n; ++len) {
        for (int i = 1; i <= n - len; ++i) {
            int j = i + len - 1;
            dp[i][j] = 1e9;

            for (int k = i; k < j; ++k) {
                int cost = dp[i][k] + dp[k + 1][j] + p[i - 1] * p[k] * p[j];
                dp[i][j] = min(dp[i][j], cost);
            }
        }
    }

    return dp[1][n - 1];
}

int main() {
    vector<int> p = {10, 20, 30, 40, 30};
    cout << "Minimum multiplications: " << matrixChainOrder(p) << endl;
    return 0;
}
```

Output:


Minimum multiplications: 30000
...Program finished with exit code 0
Press ENTER to exit console.